

HYDROPHOBIC SURFACES

LOTUS EFFECT

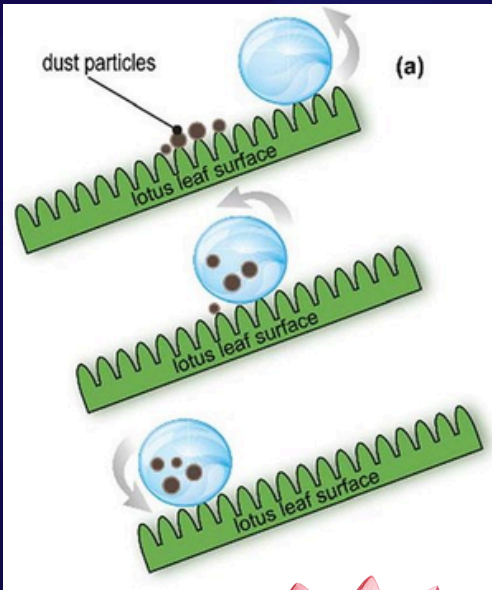
Introduction to Hydrophilia

In nanotechnology, wetting behaviour is governed by **surface free energy** at liquid-vapour interface
When studying wetting behaviour, one can consider contact angle, the interaction energy between surface and liquid:

Contact Angle (°)	Surface Classification	Example
0-30	Hydrophilic (liquid spreads across a surface)	UV-irradiated TiO ₂
30-90	Intermediate	Aluminium
≥90	Hydrophobic (contracts into discrete droplets)	Lotus leaf

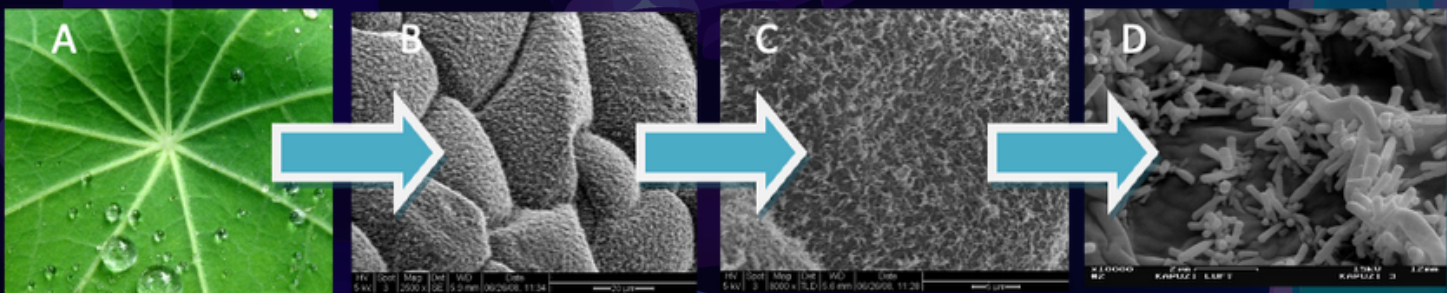
The Lotus Effect

- Lotus (Nelumbo Nucifera) is an Asian plant which is apparently “very clean”
- Superhydrophobic, due to low surface energy nanochemistry & hierarchical surface roughness
- Droplets of water on the lotus leaf appear spherical like beads
- When you splash the Lotus leaf with water, water rolls off the leaf, and in doing so drags dirt away
- The surface of the Lotus leaf **self-cleans** (“Lotus Effect”)



Nano-Structural Basis of Superhydrophobicity

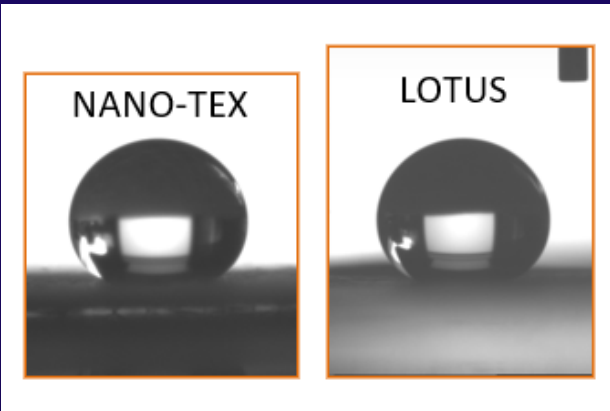
- Uses Scanning Electron Microscope (SEM) to analyse Lotus hierarchical surfaces
 - Micrometer-scale papillae
 - Nanometer-scale wax crystals (~10-100 nm)
- Trapped air pockets under droplets and greatly reduce solid-liquid contact
- Water-repellent layer, which is enhanced by the roughness of the surface, making it a superhydrophobic surface, with a contact angle of about 150deg
- As droplets come into contact with a large fraction of air, they bead up and roll off easily.



The image shows the progressive magnification of a Nasturium leaf.

Case Study I: Nano-Engineered Textiles (Nano-Tex)

Nano-Tex® textiles are **self-cleaning**, not because they need external nanometal coatings, but engineered with nano-scale “whiskers” into individual fibres with trapped air



- Repels liquid stain, dust and even liquid spill (eg Coke!)
- Pulls moisture/sweat away from skin and disperse across fabric surface
- **Quick drying, breathable & thermal comfort**



Surface Type	Contact Angle (°)
Smooth	64
Moderately porous	110
Highly porous	155
Optimised porous	167

Researched values from EU's NANOYOU

Case Study II: Porous Silicon Surfaces

- Si is intrinsically hydrophilic, but increased pore density can transition to Cassie-Baxter wetting
- High chemical stability
- Can provide durable & waterproof surfaces in **semiconducting electronics** & many tech applications to make moisture-resistant electronics

Other Applications

- Self-cleaning façade coatings
- Solar photovoltaics with less dust-induced optical loss
- Anti-fouling anti-microbial sanitary surfaces
- Water-repellent yet breathable fabrics used in protective clothing (medical) and even consumer apparel (marketed to be anti-spill)

Limitations

- Limited resistance against permanent inks & other chemicals
- Mechanical abrasion/friction can degrade nano-structures over time
- Scalability issues in nanoengineering (too small)
- Can consider research for multi-functional surfaces combining superhydrophobicity with antimicrobial or catalytic properties

References

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3148040/>
<https://www.sciencedirect.com/topics/engineering/lotus-effect>
<https://shop.nanografi.com/blog/lotus-effect-in-nanotechnology/>
<https://www.azonano.com/article.aspx?ArticleID=45>
<https://shop.nanografi.com/blog/use-of-nanotechnology-in-textile-industry-and-its-future/>
https://nanopinion.eu/files/sites/default/files/502_experiment_d1_teacher_document_11-131.pdf
<https://nanoyou.eu/en/component/content/article/87-hands-on-activities/502-experiment-with-superhydrophobic-materials.html?directory=4&Itemid=4>